Zifeng (Lauren) Liu

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EDUCATION

University of Florida Gainesville, Florida, USA

Ph.D. in Curriculum and Instruction specializing in Educational Technology *Aug. 2023 – Aug. 2026*

- Advisor: Dr. Wanli Xing
- Cumulative GPA: 3.96/4.0
- Research Assistant (Full tuition & stipends): 2023 ~ present
- Doctoral Dissertation: *Integrating Artificial Intelligence Concepts into Mathematics Learning in Virtual Schools: An Exploration of Student AI Literacy in an Online Math–AI Learning Context*
- Committee: Dr. Wanli Xing (chair), Dr. Maya Israel, Dr. Anthony Botelho, and Dr. Corinne Manley

Beijing Normal University, Beijing, China

Master of Engineering, Computer Software and Theories *Sep. 2020 – May. 2023*

- Advisor: Dr. Su Cai
- Cumulative GPA: 3.70/4.0
- Master's thesis: *Facial Expression Recognition and Application based on Contrastive Learning and Visual Attention*

Beijing Technology and Business University, Beijing, China

Bachelor of Engineering, Computer Science and Technology *Sep. 2016 – May. 2020*

- Advisor: Dr. Zhongming Han and Dr. Yi Chen
- Cumulative GPA: 4.27/5.0 (ranked number 1 among 58 students)
- Bachelor thesis: *Multi-User and Multi-Dimensional Interactive Data Visualization*

RESEARCH INTERESTS

- AI, Computing Education
- Educational Data Mining
- Learning Analytics
- STEM-integrated Computing Education
- Fair, accountable, and transparent (FAccT) AI

AWARDS & HONORS

1. AERA Online Teaching and Learning SIG Best Paper Award (2026)
2. Elected Full Member, Sigma Xi, The Scientific Research Honor Society (2026)
3. Vernice Law Hearn Scholarship for Outstanding Graduate Students, University of Florida (2025, $2,000.00)
4. Best Full Paper Award, the 25th IEEE International Conference on Advanced Learning Technologies (ICALT 2025)
5. ACM International Computing Education Research (ICER) Conference Travel Grant (2025, $500)
6. First Prize of Excellent Graduate Student Scholarship, Beijing Normal University (2021, 2022, ¥12,000.00)
7. Best Paper Award Nomination, the 28th International Conference on Computers in Education (ICCE 2020)
8. Outstanding Freshman Scholarship, Beijing Normal University (2020, ¥10,000.00)

9. Excellent Individual of Summer Volunteer Teaching Program, Beijing Normal University (2022)
10. Excellent Graduate of Beijing, Beijing Municipal Education Commission (2020, top 5%)
11. National Scholarship of China, Ministry of Education of China (2018-2019, top 0.2%, ¥8000.00)
12. Dean's Scholarship, Beijing Technology and Business University (2018-2019, top 0.3%, ¥1,0000.00)
13. Outstanding Student Scholarship, Beijing Technology and Business University (2018, 2019)
14. Student LeaderShip Award, Beijing Technology and Business University (2017-2018)
15. National Scholarship for Encouragement of China, Ministry of Education of China (2016, 2017, ¥5000.00)

PUBLICATIONS

Refereed Journal Articles

1. **Liu, Z.**, Xing, W., Monteith, B., Pei, B., & Zhang, Y. (accepted). Engaging secondary students in data science learning through a mathematical logic approach: Insights from surveys and log data. Manuscript submitted for publication to *ACM Transactions on Computing Education*. [Impact Factor: 3.8]
2. **Liu, Z.**, Xing, W., Jiang, Y., Li, C., Kim, T., & Li, H. (2025). Leveraging contrastive learning to improve group and individual fairness in predictive analytics for online learning. *Journal of Computing in Higher Education*, *37*, 1341–1370. https://doi.org/10.1007/s12528-025-09468-y [Impact Factor: 4.9]
3. **Liu, Z.**, Xing, W., Jiao, X., & Li, C. (2025). Exploring Fairness and Explainability in LLM-Generated Support for Online Learning Discussion Forums. *Journal of Learning Analytics*, 1-26. https://doi.org/10.18608/jla.2025.8885 [Impact Factor: 3.6]
4. **Liu, Z.**, Xing, W., Li, C., Zhang, F., Li, H., & Minces, V. (2025). Exploring Automated Assessment of Primary Students' Creativity in a Flow-Based Music Programming Environment. *Journal of Learning Analytics, 12*(2), 83-104. https://doi.org/10.18608/jla.2025.8835. [Impact Factor: 3.6]
5. **Liu, Z.**, Xing, W., Jiao, X., & Li, C. What are the differences between student and ChatGPT-generated pseudocode? Detecting AI-generated pseudocode in high school programming using explainable machine learning. *Education and Information Technologies* (2025). https://doi.org/10.1007/s10639-025-13385-z. [Impact Factor: 5.4]
6. **Liu, Z.**, Xing, W., Ngo, B., Jiao, X. et al.. (2025). Engagement patterns of middle school students with AI teachable agents in mathematics learning. *Scientific Reports, 15*, 40971. https://doi.org/10.1038/s41598-025-24841-8 [Impact Factor: 3.9]
7. **Liu, Z.**, Cheon, S., Stanbury, A., Jiao, X., Xing, W., & Kang, H. (2025). Towards contextual-based AI: A scoping review of artificial intelligence in X reality for personalized learning. Computers and Education: Artificial Intelligence. https://doi.org/10.1016/j.caeai.2025.100523. [Impact Factor: 24.0]
8. Xing, W., Song, Y., **Liu, Z.**, Du, H., Zhu, W., & Li, C. (accepted). Investigating Expert and Peer Tutoring Behaviors in a Large Online Discussion Forum Using Temporal Dynamic Analytics. Manuscript submitted for publication to *Journal of Learning Analytics*.
9. Xing, W., **Liu, Z.**, Kim, T & Song, Y.,. Why do students leave instructional videos: understanding students' in-video dropout behavior in a large online math learning platform? Manuscript submitted for publication to *Distance Education*. https://doi.org/10.1080/01587919.2025.2579792 [Impact Factor: 3.0]
10. Song, Y., Kim, J., **Liu, Z.**, Li, C., & Xing, W. (2025). Students' perceived roles, opportunities, and challenges of a generative AI-powered teachable agent: a case of middle school math class. *Journal of Research on Technology in Education*, 1–19. https://doi.org/10.1080/15391523.2024.2447727. [Impact Factor: 5.0]
11. Owoputi, R., Kwok, A., **Liu, Z.**, Zhu, W., Xing, W., & Ray, S. (in press). *Immersive virtual environment for automotive security education*. *IEEE Transactions on Education.* [Impact Factor: 2.0]
12. Xing, W., Song, Y., Li, C., **Liu, Z.**, Zhu, W., Oh, H. (2025). Development of a generative AI-powered teachable agent for middle school mathematics learning: a design-based research study. *British Journal of Educational Technology.* 2043-2077. https://doi.org/10.1111/bjet.13586 [Impact Factor: 8.1]
13. Song, Y., Kim, J., Xing, W., **Liu, Z.**, Li, C., & Oh, H. (2025). Elementary school students' and teachers' perceptions toward creative mathematical writing with Generative AI. *Journal of Research on Technology in Education*, 1–23. https://doi.org/10.1080/15391523.2025.2455057. [Impact Factor: 5.0]
14. Xing, W., Fang, Z., Zhang, H., Kamiyama, T., **Liu, Z.**, & Kim, T. (2025). Making the 'mathematics register' accessible to students: an exploratory study of two teachers' discourse in an online lesson on polynomial

expressions. *Language and Education*, 1–24. https://doi.org/10.1080/09500782.2025.2542861 [Impact Factor: 2.8]

15. Li, H., Xing, W., Zhu, W., Zhang, S., & **Liu, Z.** (2025). Should educational AI models include gender attribute? Explaining the why based on environmental psychology course with gender imbalance. *Journal of Computing in Higher Education*. https://doi.org/10.1007/s12528-025-09467-z [Impact Factor: 4.9]
16. Zhu, W., Xing, W., Kim, E. M., Li, C., Wang, Y., Yang, Y., & **Liu, Z.** (2025). Integrating image-generative AI into conceptual design in computer-aided design education: Exploring student perceptions, prompt behaviors, and artifact creativity. *Educational Technology & Society, 28*(3), 166-183. https://doi.org/10.30191/ETS.202507_28(3).SP11. [Impact Factor: 6.0]
17. Jiang, Y., Fan, Y., & **Liu, Z.** (2026). Generative AI in Art Education: A Systematic Review of Research Trends, Tool Applications, and Outcomes (2019–2025). *Education Sciences, 16*(1), 47. https://doi.org/10.3390/educsci16010047 [Impact Factor: 2.6]
18. Cai, S., **Liu, Z.**, Liu, C., & others. (2022). Effects of a BCI-based AR inquiring tool on primary students' science learning: A quasi-experimental field study. *Journal of Science Education and Technology, 31,* 767–782. https://doi.org/10.1007/s10956-022-09991-y. [Impact Factor: 5.5]
19. Liu, E., Cai, S., **Liu, Z.**, & Liu, C. (2023). WebART: Web-based augmented reality learning resources authoring tool and its user experience study among teachers. *IEEE Transactions on Learning Technologies, 16*(1), 53–65. https://doi.org/10.1109/TLT.2022.3214854. [Impact Factor: 4.9]

Manuscripts Under Review and In Progress

1. **Liu, Z.**, Xing, W., Zhang. S., Israel, M., & Minces, V. (under review). Connecting Music and Elementary Computing Education through Flow-Based Programming: Insights from Teachers and Students. Manuscript prepared for submitting to *Computer Science Education*.
2. Jiao, X., **Liu, Z.**, & Cai, S. (under review). Enhancing Collaboration in AR-based Inquiry: Effects of Structured Collaborative Scripts on Students' Performance and Group Dynamics. Manuscript submitted for publication to *Interactive Learning Environments*.
3. Xing, W., Fang, Z., **Liu, Z.**, & Guo, R. (under review). *How teachers talk algebra online: Linking discourse patterns, instructional experiences, and student outcomes*. Manuscript submitted for publication to *Journal of Learning Analytics*.
4. Zhang, F., Xing. W., & **Liu, Z.**. (2026). SAGE-AAC: A retrieval-first, safety-governed multi-agent pipeline for explainable text-to-pictogram generation in inclusive AAC-supported learning. *British Journal of Educational Technology*. Manuscript submitted for publication.
5. Li, H., Xing, W., Lyu, B., Li, C., & **Liu, Z.** (2026). Engaging the mathematics learner through teachable AI agents: How visual backgrounds and talking-head videos shape student perceptions, dialogue behaviors, and learning experiences. *British Journal of Educational Technology*. Manuscript submitted for publication.
6. Liu, T., Du, H., **Liu, Z.**, & Pei, B. (2026). AI delegation across cognitive responsibilities: Patterns and associations with performance in statistical reasoning tasks. *Journal of Educational Computing Research*. Manuscript submitted for publication.
7. Guo, R., **Liu, Z.**, Thieren, S., Xing, W., & Fang, Z. (2026). Unpacking teacher talk in video lectures: Metadiscourse, experience/expertise, and their effects on student engagement and learning. *Journal of Computer Assisted Learning*. Manuscript submitted for publication.

Refereed Conference Proceedings

1. **Liu, Z.**, Mondol, A., Jiao, X., Chao, J., Li, L., & Xing, W. (in press). *From examples to rules? Exploring inductive reverse engineering and deductive few-shot coding via LLMs for qualitative data analysis*. In Proceedings of the Artificial Intelligence in Education (AIED) 2026 Conference. [≈16% acceptance rate]
2. **Liu, Z.**, Li, H., Chao, J., & Xing, W. (2026, Jan.). *Brains vs. Algorithms? How experts and students see AI-generated distractors.* In *Proceedings of the 2026 AAAI Symposium on Educational Advances in Artificial Intelligence (AAAI-26).* [≈30% acceptance rate]
3. **Liu, Z.**, Li, L., Chao, J., Mondol, A., Xing, W., & Zhang, Y. (2026, Mar.). *Do all roads lead to AI literacy? Clustering behavioral patterns and examining outcomes in an online AI literacy module for secondary school students*. In *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK '26)*.

ACM. [≈30% acceptance rate]

4. **Liu, Z.**, Xing, W., & Fang, Z. (2026, Mar.). *Talking the Talk: Linking Instructional Discourse Patterns to Student In-video Dropout and Learning Outcome*. In *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK '26)*. ACM. [≈30% acceptance rate]
5. Kim, T., **Liu, Z.**, & Xing, W. (2026). Learning in VR with and without paper: An entropy-based log analysis. In *2026 IEEE 26th International Conference on Advanced Learning Technologies (ICALT)* (pp. xx–xx). IEEE. **[Best Paper Nomination]**
6. Jiao, X., & **Liu, Z.**. (2026). Unveiling Role Dynamics in AR-Based Collaborative Inquiry. In *Proceedings of the 2026 International Conference on Computer-Supported Collaborative Learning (CSCL 2026)*. International Society of the Learning Sciences.
7. Li, H., **Liu, Z.**, Chao, J., & Xing, W. (2026). Theoretical construction and case study of temporal learning analytics based on learning engagement dynamics framework. In *Proceedings of the International Conference of the Learning Sciences (ICLS 2026)*. International Society of the Learning Sciences. [≈30% acceptance rate]
8. Kim, T., **Liu, Z.**, Xing, W., & Roy, T. (2026). Does More Exploration Mean Better Learning? Rethinking VR Log Data Through Instructional Sequences. In *Proceedings of the International Conference of the Learning Sciences (ICLS 2026)*. International Society of the Learning Sciences. **[Best Paper Nomination]**
9. Jiao, X., **Liu, Z.**, Mei, S., & Cai, S. (2025). Beyond the screen: Enhancing augmented reality collaborative inquiry with social scripts. In *Proceedings of the 16th International Learning Analytics and Knowledge Conference (LAK '26)*. ACM. [≈30% acceptance rate]
10. Du, H., **Liu, Z**., Xing, W., & Zhang, Y. (2025, Nov.). A logic-programming-based CS professional development for secondary teachers. Paper accepted for presentation at the 14th International Conference on Educational Innovation through Technology (EITT 2025), Tokyo City University, Yokohama, Japan.
11. Li, H., Xing, W., Li, C., & **Liu, Z**. (2025, December) Regularizing chart-specialized vision-language models for K-12 hand-drawn math VQA: Performance and representation effects. In *Proceedings of the 14th International Conference on Computational Data and Social Networks (CSoNet)*.
12. **Liu, Z.**, Ganapathy Prasad, P., Ngo, B., Jiao, X., & Xing, W. (2025). *A human–AI collaborative assessment of AI-generated vs. human-created MCQ distractors*. In *Proceedings of the 29th International Conference on Computers in Education (ICCE 2025)*, Chennai, India.
13. Kim, T., **Liu, Z.**, Xing, W., Li, H., & Oh, H. (2025, Jun.). Emotional dynamics in asynchronous math discussions: Analyzing the impact of negative emotions on learning outcomes. *In Proceedings of the 2025 International Conference of the Learning Sciences (ICLS 2025).* International Society of the Learning Sciences (ISLS). June 10–13, Finland. https://repository.isls.org//handle/1/11805 [≈30% acceptance rate]
14. Li, H., Xing, W., Zhu, W., Li, C., Lyu, B., **Liu, Z**., & Heffernan, N. (accepted). Leveraging multi-modality and collaborative filtering for supporting automatic scoring in mathematics education. In *Proceedings of the 26th International Conference on Artificial Intelligence in Education.* vol 2591. Springer, Cham. https://doi.org/10.1007/978-3-031-99264-3_39 [≈30% acceptance rate]
15. Li, H., Xing, W., Lyu, B., Zhu, W., **Liu, Z.**, & Li, H. (accepted). An automated aesthetic assessment framework of mathematical story images validated by click counts. In Proceedings of the 18th ACM Conference on Learning@ Scale. https://doi.org/10.1145/3698205.3733923 [≈30% acceptance rate]
16. Jiao, X., Huang, H., **Liu, Z.**, Cai, S., & Fan, Z. (2025). Beyond the screen: Enhancing augmented reality collaborative inquiry with social scripts. In *Proceedings of the 25th IEEE International Conference on Advanced Learning Technologies (ICALT 2025)*, Changhua, Taiwan. IEEE. **[Best Full Paper Award]**
17. **Liu, Z.**, Song, Y., Yang, Q., Xing, W., & Guo, J. (2025, Jun.). Exploring the Impact of a Simulation-Based Learning Tool on Undergraduate Quantum Computing Education. In *2025 ASEE Annual Conference & Exposition*.
18. **Liu, Z.**, Xing, W., Zhang, F., & Li, C. (2025, March). A visual programming approach to enhance spatial computational thinking skills in upper-elementary students [Poster]. The 15th International Learning Analytics & Knowledge Conference, Dublin, Ireland.
19. **Liu, Z.**, Xing, W., Israel, M., & Minces, V. (2025). Enhancing Pre-Service Teacher Confidence And Engagement Through Flow-Based Music Programming. Proceedings of EDULEARN25 (pp. 4127–4133). IATED Academy. https://doi.org/10.21125/edulearn.2025.1070
20. Oh, H., **Liu, Z.**, & Xing, W. (2025, March 3–7). Do actions speak louder than words? Unveiling linguistic

patterns in online learning communities using cross recurrence quantification analysis. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference (LAK '25)* (pp. 992–998). Association for Computing Machinery. https://doi.org/10.1145/3706468.3706569 [≈30% acceptance rate]

21. Li, H., Xing, W., Li, C., Zhu, W., Lyu, B., Zhang, F., & **Liu, Z.** (2025, March). Who Should Be My Tutor? Analyzing the Interactive Effects of Automated Text Personality Styles Between Middle School Students and a Mathematics Chatbot. In *Proceedings of the 15th Learning Analytics and Knowledge Conference* (LAK2025) (pp. 1–7). https://doi.org/10.1145/3706468.3706537 [≈30% acceptance rate]
22. **Liu, Z.**, Zhang, S., Israel, M., Smith, R., Xing, W., & Minces, V. (2025, Feb). Engaging K-12 students with flow-based music programming: An experience report on its impact on teaching and learning. *In Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE TS 2025)*, February 26–March 1, 2025, Pittsburgh, PA, USA. https://doi.org/10.1145/3641554.3701902 [≈30% acceptance rate]
23. **Liu, Z.**, Jiao, X., Xing, W., & Zhu, W. (2025, Feb). Detecting AI-generated pseudocode in high school online programming courses using an explainable approach. *In Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE TS 2025)*, February 26–March 1, 2025, Pittsburgh, PA, USA. https://doi.org/10.1145/3641554.3701942 [≈30% acceptance rate]
24. **Liu, Z.**, Guo, R, Song, Y., Xing, W. (2024). WIP: understanding students' in-video dropout behavior in large online math learning platform. In *Proceedings of 2024 IEEE Frontiers in Education*, Oct 13–16, 2024, Washington, D.C., USA.
25. **Liu, Z.**, Guo, R., Jiao, X., Gao, X., Oh, H., & Xing, W. (2024, June). How AI Assisted K-12 Computer Science Education: A Systematic Review. In *2024 ASEE Annual Conference & Exposition*.
26. Oh, H., Guo, R., Xing, W., **Liu, Z.**, Song, Y., & Li, C. (2024, June). The Seamless Integration of Machine Learning Education into High School Mathematics Classrooms. In *2024 ASEE Annual Conference & Exposition*.
27. Jiao, X., **Liu, Z.**, Zhou, H., & Cai, S. (2022, July). The Effect of Role Assignment on Students' Collaborative Inquiry-based Learning in Augmented Reality Environment. In *2022 International Conference on Advanced Learning Technologies (ICALT)* (pp. 349-351). IEEE.
28. Feng, Z., Gong, C., Jiao, X., **Liu, Z.**, & Cai, S. (2022, July). The Effects of AR Learning Environment to Preschool Children's Numerical Cognition. In *2022 International Conference on Advanced Learning Technologies (ICALT)* (pp. 352-356). IEEE.
29. **Liu, Z.**, Jiao, X., & Cai, S. (2021, April 4). *Effects of augmented reality on students' online physics learning*. Paper presented at the 2021 Annual Meeting of the American Educational Research Association (AERA), Virtual Conference. Retrieved August 25, 2022, from the AERA Online Paper Repository.
30. Jiao, X., **Liu, Z.**, & Cai, S. (2020, November 23–27). *Impact of embedded cognitive scaffolding of augmented reality technology on elementary school students' science learning*. Paper presented at the 28th International Conference on Computers in Education (ICCE 2020), Virtual Conference. [**Best Paper Nomination**]

Poster and Workshop Papers

31. **Liu, Z.**, Jiang, Y., & Xing, W. (2026, Feb.). *Exploring the use of LLMs for assessing creativity in student programming artifacts.* In *Proceedings of the 57th ACM Technical Symposium on Computer Science Education (SIGCSE TS 2026)* ACM.
32. **Liu, Z.**, Jiao, X., Li, C., & Xing, W. (2024, July). Fair Prediction of Students' Summative Performance Changes Using Online Learning Behavior Data. In *Proceedings of the 17th International Conference on Educational Data Mining* (pp. 686-691).
33. **Liu, Z.**, Xing, W., & Li, C. (2024, July). Explainable analysis of AI-generated responses in online learning discussions. In *Educational Data Mining 2024 Workshop: Leveraging Large Language Models for Next-Generation Educational Technologies*. https://doi.org/10.13140/RG.2.2.24309.38881
34. **Liu, Z.**, Zhang, S., Xing, W., Minces, V., Israel, & Barron, A. (2025, June). A NSF ITEST Program: Integrating Music and Flow-Based Programming Builds Teachers' Confidence in Computer Science. In *2025 ASEE Annual Conference & Exposition*.

Patent

1. Cai, S., **Liu, Z.**, Changhao Liu, & Haitao Zhou. (China, 2021). *A non-invasive brain-computer interface-based*

attention feedback method (Patent No. ZL 2021-1-1283053.5).
2. Cai, S., **Liu, Z.**, & Zhang, Y.. (2023, under review). *A grid-based self-attention facial expression recognition method using supervised contrastive learning*. (China, Patent pending).

Presentations

1. **Liu, Z.,** Xing, W., & Minces, V. (2026, July 13–15). *A review of current music and programming platforms for classroom teaching* [60-minute breakout session]. Computer Science Teachers Association Annual Conference (CSTA 2026), New Orleans, LA, United States.
2. **Liu, Z.**, & Xing, W. (2026, April). *Exploring the use of LLMs for assessing creativity in student programming artifacts.* Paper presented at the 2026 American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA.
3. Xing, W., & **Liu, Z.** (2026, April). *Exploring expert and peer tutoring dynamics in online learning discussions: A temporal learning analytics approach.* Paper presented at the 2026 American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA.
4. Jiao, X., **Liu, Z.**, & Cai, S. (2026, April). *The role of group interaction patterns in shaping learning outcomes in AR-enhanced collaborative inquiry.* Paper presented at the 2026 American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA.
5. Kim, T., **Liu, Z.**, Xing, W., & Roy, T. (2026, April). *One log, two stories: How learning order shapes the interpretation of VR log events.* Paper presented at the 2026 American Educational Research Association (AERA) Annual Meeting, Los Angeles, CA.
6. Zhang, S., Ganapathy, P. P., Earle-Randell, T. V., **Liu, Z**., Shi, Y., Bhat, S., & Israel, M. (2026, April). Analyzing High School Students' Code Comprehension and Strategies Using SOLO Taxonomy and Large Language Models. Poster presented at the Annual Meeting of the American Educational Research Association (AERA), Los Angeles, CA, United States.
7. Fang, Z., Xing, W., **Liu, Z.**, & Guo, R. (2026, April 18–19). *Teaching algebra online: Patterns of teacher talk and their relationship to student outcomes* [Oral presentation]. Annual Conference of the Association for Applied Corpus Linguistics (AACL 2026), Gainesville, FL, United States.
8. Jiang, Y., & **Liu, Z**. (2026, January 3-7). Generative artificial intelligence in art education: An overview of current research themes and future research directions [Paper presentation]. The IAFOR International Conference on Education in Hawaii (IICE2026), Honolulu, HI, United States.
9. **Liu, Z**., Ngo, B., & Xing, W. (2025). Evaluating AI-generated distractors in programming education: A human-AI collaborative approach [Poster Presentation]. In Proceedings of the 2025 ACM Conference on International Computing Education Research (ICER 2025). ACM.
10. **Liu, Z.,** Xing, W., Zhang, F., & Li, C**.** (2025, November). *Assessing creativity in music programming: Can AI automate the process?* Concurrent presentation at the AECT International Convention, Las Vegas, NV, United States.
11. Xing, W., **Liu, Z.**, & Zhang, F. (2025, November). *Enhancing spatial computational thinking in upper-elementary students through visual programming* [Poster presentation]. AECT International Convention, Las Vegas, NV, United States.
12. **Liu, Z.**, & Xing, W. (2025). *M-Flow: A new flow-based music programming platform* [Demo]. *Warren B. Nelms Annual IoT Conference 2025*, Gainesville, FL, United States.
13. Jiang, Y., Fan, Y., & **Liu, Z**. (2026, January 3-7*). Generative artificial intelligence in art education: An overview of current research themes and future research directions* [Paper presentation]. The IAFOR International Conference on Education in Hawaii (IICE2026), Honolulu, HI, United States.
14. **Liu, Z**., Jiang, Y., Li, C., & Xing, W. (2025). Using supervised contrastive learning for improving individual fairness in AI online performance prediction. Presented at the 2025 American Educational Research Association (AERA) Annual Meeting, Denver, Colorado.
15. **Liu, Z.**, Song, Y., & Xing, W. (2025). Identifying factors of in-video dropout in middle school online math learning using survival analysis. Presented at the 2025 American Educational Research Association (AERA) Annual Meeting, Denver, Colorado.
16. **Liu, Z.**, Monteith, B., Chao, J., Wiedemann, K., Fofang, J. B., Li, L., Ma, D., Mohamed, R., Mondol, A., Jo, Y., Fleetwood, A., Lipien, L., Zhang, Y., & Xing, W. (2025, February 17–19). Using entropy analysis to explore

student engagement in an online high school data science course. Presented at *DSE-K12 Conference 2025*, Hyatt Regency San Antonio Riverwalk, San Antonio, TX, USA.

17. Xing, W., **Liu, Z.**, & Song, Y. (2025). Teaching quantum computing to undergraduate students using a multimedia and simulation-based learning technology. Presented at the 2025 American Educational Research Association (AERA) Annual Meeting, Denver, Colorado.
18. Xing, W., Song, Y., Li, C., **Liu, Z**. (2025) Flipping the Roles: Engaging Middle School Students in Learning-by-teaching with a Generative AI-powered Teachable Agent. Presented at 2025 American Educational Research Association (AERA). Denver, Colorado.
19. Kim, T., Xing, W., **Liu, Z**., & Li, H. (2025). Investigating the Impact of Negative Emotions in High School Students' Asynchronous Online Math Discussions. Presented at the 2025 American Educational Research Association (AERA) Annual Meeting, Denver, Colorado.
20. **Liu, Z.**, Xing, W. , & Zhu, W.(2023). How AI assisted k-12 computer science education? A systematic review. [Symposium]. University of Notre Dame, US.
21. **Liu, Z.**, Xing, W. , & Zhu, W.(2023). Integrating Generative AI in Augmented Reality: A New Paradigm for Educational Resource Creation. [Symposium]. University of Notre Dame, US.
22. Oh, H., Guo, R., Xing, W., Song, Y., Li, C., **Liu, Z**. (2024, October 19-23). Integrating Machine Learning into High School Mathematics [Paper presentation]. Association for Educational Communications & Technology International Convention 2024, Kansas City, MO, United States.
23. Xing, W., **Liu, Z**., & Fang, Z. (2024,October 19-23). An exploratory study of mathematics teachers' discourse in an online lesson on polynomial expressions [Paper presentation]. Association for Educational Communications & Technology International Convention 2024, Kansas City, MO, United States.
24. Xing, W., **Liu, Z.**, & Song, Y. (2024,October 19-23). Understanding in-video dropout behavior: Analyzing student engagement in online math education [Paper presentation]. Association for Educational Communications & Technology International Convention 2024, Kansas City, MO, United States.
25. **Liu, Z.**, Song, Y., Xing, W., Guo, J., Feng, P., & Kim, G. (2024). Enhancing students' quantum computing learning through multimedia and simulation-based tools. Proposal submitted for presentation at the AECT International Convention, Kansas City, MO, October 19–23.
26. **Liu, Z.**, Jiang, Y., Li, C., & Xing, W. (2024, October 19–23). Improve individual fairness in online learning performance prediction [Paper presentation]. Association for Educational Communications & Technology International Convention 2024, Kansas City, MO, United States.
27. Fang, Z., Xing, W., & **Liu, Z**. (2024, December 4-7). Mathematics teachers' discourse in an online lesson on polynomial expressions. Presented at the Literacy Research Association's 74th Annual Conference, Westin Peachtree Plaza, Atlanta, GA, USA.
28. **Liu, Z**., Xing, W., Ray, S., & Owoputi, R. (2024, December 5–6). Enhancing automotive cybersecurity education in higher education through an immersive virtual environment [Poster presentation]. Warren B. Nelms Annual IoT Conference 2024, Gainesville, FL, United States.
29. Li, H., **Liu, Z**., & Xing, W. (2024, December 5–6). Leveraging secure IoT for intelligent multi-modal assessment in mathematics education [Poster presentation]. Warren B. Nelms Annual IoT Conference 2024, Gainesville, FL, United States.
30. **Liu, Z.**, Xing, W., Zhang, F., & Li, C. (2025, March). A visual programming approach to enhance spatial computational thinking skills in upper-elementary students [Poster presentation]. The 15th International Learning Analytics & Knowledge Conference, Dublin, Ireland.

Invited Talk & Presentation

1. **Liu, Z. (2026, April 11).** *Human–AI collaboration in educational assessment: Generative AI for distractors and feedback* [Invited talk]. Cohere Labs Events.

Doctoral Symposium

1. *An Explainable Model for AI-Generated Pseudocode Detection in Online High School Programming Courses*, IEEE Frontiers in Education (FIE) 2024, Washington, D.C.
2. *Automatic Distractor and Feedback Generation in Online AI Education: A Design-Based Research Study*, ACM International Computing Education Research (ICER) 2025, Virginia

CURRENT RESEARCH & GRANT EXPERIENCE

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1. AI4VS: AI Across the Curriculum for Virtual Schools *Researcher Aug. 2024 - present*
(Funded by DOE EIR #S411C230070, $3,999,322; PI: Dr. Jie Chao (Concord), Co-PI: Dr. Wanli Xing)
- Led research on using generative AI to support pedagogical multiple-choice question development, including distractor generation and automated feedback.
- Led research on modeling students' learning profiles and interaction patterns within online AI learning modules.
- Co-designed an AI-integrated mathematics curriculum for high school students in virtual learning environments.
- Developed measurable learning objectives and activity sequences aligned with AI "big ideas" and relevant mathematics standards.
- Designed and iteratively refined user interface components; coordinated user testing and synthesized usability findings to inform revisions.

2. M-Flow: Using Flow-Based Music Programming to Engage Children in Computer Science
(Funded by NSF iTEST #2241715, $1,227,507; PI: Dr. Wanli Xing) *Researcher Sep. 2023 – present*
- Led development of the flow-based music programming platform (M-Flow), including feature implementation and iterative refinement.
- Led the design and development of research instruments for teachers and students (e.g., surveys, interview protocols, and learning measures).
- Led professional development interviews with 7 elementary school teachers and supported qualitative data preparation for analysis.
- Led survey and platform log-data analyses to evaluate the efficacy of the M-Flow platform and associated curriculum.

3. eSPAC3: A Culturally Relevant Approach to Spatial Computational Thinking Skills and Career Awareness through an Immersive Virtual Environment
(Funded by NSF iTEST, $227,619. PI: Dr. Wanli Xing) *Researcher* Sep. 2023 – *present*
- Led the design and development of survey instruments for peer mentors, parents, and students.
- Conducted pre–post survey analyses assessing spatial computational thinking and learner self-efficacy.
- Contributed to drafting of the annual technical report, including research outcomes and data analysis summaries.

4. ALTER-Math: AI-augmented Learning by Teaching to Enhance and Renovate Math Learning
(Funded by Learning Engineering Virtual Institute (LEVI), Schmidt Future Foundation, PI: Dr. Wanli Xing)
Researcher Oct. 2023 – present
- Led log-data analyses of students' interactions with an AI teachable agent to characterize learning behaviors.
- Designed student survey measures related to AI literacy and mathematics learning motivation.
- Contributed to the Spring 2024 classroom study at P.K. Yonge Developmental Research School, including implementation support and data collection.

5. ART-Math: A Student-centered Interactive Mathematical Learning and Creation Platform powered by AI
(Funded by Department of Education #91990023 C0022, $4,000,000, PI: Dr. Wanli Xing)
Researcher Apr. 2024 – Apr. 2025
- Contributed to organizing a classroom study to examine the initial efficacy of the ART-Math platform.
- Contributed to usability interviews with approximately 20 elementary school students and 5 teachers across the

United States; supported synthesis of interview insights for design iteration.

6. LogicDS: A Logic Programming Approach to Integrate Computing with Middle School Science Education
(Funded by NSF #1901704, $421,755; PI: Dr. Yuanlin Zhang, Co-PI: Dr. Wanli Xing)
Researcher Apr. 2024 – Apr. 2025
▪ Analyzed student learning log data using entropy-based measures and examined associations with student performance outcomes.

7. ML4Math: Integrating Machine Learning and Mathematics Education Using a Concreteness Fading Approach
(Funded by ONR: #N00014-23-S-F001 , $600,000 ; PI: Dr. Wanli Xing)
Researcher Apr. 2024 – Sep. 2025
▪ Assisted with Fall 2025 classroom implementation planning and data collection at Loften High School (Gainesville).
▪ Co-designed and revised course materials to support instruction and classroom deployment.

8. INQUIRE: Innovating Quantum-Inspired Learning for Undergraduates in Research and Engineering
(Funded by NSF IUSE program #2142552, $1,250,000, PI: Dr. Gloria Kim, Co-PI: Dr. Wanli Xing)
Researcher Nov. 2023 — Apr. 2024
▪ Contributed to the 2024 classroom implementation of the Spin-Qubit Lab at the University of Florida, including support for instructional activities and study logistics.

RESEARCH GRANT WRITING

1. Contributed to the writing of the research grant proposal "ML4Math: Integrating Machine Learning and Mathematics Education Using a Concreteness Fading Approach", successfully funded by the U.S. Department of Navy, Office of Naval Research (ONR), (#N00014-23-S-F001; $600,000). PI: Dr. Wanli Xing.
2. Contributed to the writing of the research grant proposal "Magical-Math: Multimodal Artificial Generative Intelligence Championing Advanced Learning Sciences for Math Education", submitted to Google's General Research Grant Program ($1,500,000). PI: Dr. Wanli Xing.
3. Contributed to the research grant proposal "REU Site: Secure, Accessible, and Sustainable Transportation", submitted to the NSF REU Site program ($400,000). PI: Dr. Wanli Xing.
4. Contributed to the preparation of the research grant proposal "Adaptive Contextual Learning in Microelectronics through Immersive Virtual Reality", submitted to NSF's Research on Innovative Technologies for Enhanced Learning (RITEL) program ($900,000). PI: Dr. Hyo Kang.
5. Contributed to the preparation of the research grant proposal "Collaborative Proposal: CyberAI Innovation: Secure Hardware and Integrated Education for Learning and Defense in Artificial Intelligence (SHIELD-AI)", submitted to NSF's CyberAICorps Scholarship for Service (CyberAI SFS) program ($500,000). PI: Dr. Wanli Xing.
6. American Educational Research Association (AERA) Dissertation Grant (Under Review). Investigating the Effects of Integrating Artificial Intelligence Concepts into Mathematics Learning in Virtual Schools ($27,500). PI: Zifeng Liu.

TEACHING EXPERIENCE

ML4Math: Generative AI and Mathematics Learning (High School Level), Loften High School, Florida
Teaching Assistant *July. 2025*
▪ Assisted in designing and delivering lessons for 50 students on AI ethics, responsible AI use, and critical evaluation of AI-generated content.
▪ Facilitated student activities involving generative AI tools for text, images, and videos

AI4VS: AI in Math Module (Online, High School Level), Florida Virtual School & The Concord Consortium
Teaching Assistant *Feb. 2025– Mar. 2025*
▪ Co-designed AI-integrated mathematics curriculum within a virtual learning environment for 110 high school students.

▪ Developed learning objectives and instructional activities aligned with AI Big Ideas and state mathematics standards.
▪ Designed user interface components for the virtual learning platform and conducted formative user testing.
▪ Provided feedback on students' online learning responses and supported them by answering content-related questions.

EDG 6931-Generative Artificial Intelligence in Education (Online Graduate Level), University of Florida
Teaching Assistant *Jan. 2025– May. 2025*
▪ Designed and led the discussion activities (key AI in Education concepts and conceptual frameworks).
▪ Graded students' assignments and gave feedback to team projects.
▪ Managed the course materials (readings, videos etc.), assignments, and online discussions on Canvas.

ART-Math: Creative Math Writing with Generative AI (Elementary Level), P.K.Yonge Developmental Research Elementary school
Instructor *Feb. 2024 - Mar. 2024*
▪ Instructional design, teaching of a total of 48 students in two formal math classes, learning artifact evaluation.

Object-Oriented Programming (Java, Undergraduate Level), Beijing Normal University
Assistant Instructor, *Mar. 2022 – July. 2022*
▪ Provided instruction and hands-on support for 38 students during laboratory sessions including debugging and Q&A.
▪ Developed instructional and assessment examples to support the teaching of object-oriented programming concepts in Java.
▪ Designed problem-based learning projects and offered them as options for students' final project tasks.

VR/AR Scenario Design (Undergraduate Level), Beijing Normal University
Teaching Assistant, *Sep. 2021 – Dec. 2021*
▪ Taught AR development techniques using C# and Vuforia, including marker tracking and interaction design.
▪ Reviewed 20 students' projects and provided detailed feedback on functionality usability and debugging issues.

Mathematics Teaching Assistant (Middle School Level), Huayuan School, Laiwu District, Jinan, Shandong Province, China
Teaching Assistant *Jun*. 4 2021 & *Dec*. 29-30 2021
▪ Assisted in delivering middle school mathematics lessons on relative motion problems (catch-up and meeting problems) for two classes (N = 80).
▪ Provided in-class technical support, including testing classroom equipment and troubleshooting instructional technologies.
▪ Responded to students' questions during class and supported learning activities under the instructor's guidance.
▪ Assisted in designing instructional slides and supported the development of augmented reality (AR)–based learning materials for mathematics instruction.

Teaching Assistant (Middle School Level), Xihu Experimental Middle School, Hainan Province, China
Teaching Assistant *Oct.* 22 - 24, 2021
▪ Assisted in delivering a science lesson on the additive color model of light (RGB) using an experimental–control instructional design.
▪ Contributed to the development of AR-supported instructional materials and corresponding slides to facilitate students' understanding of color mixing and light properties while ensuring content equivalence across groups.

Excellent Individual of Summer Volunteer Teaching Program (Online, Middle School Level), Beijing Normal University
Instructor *July. 2021 - August. 2021*
▪ Taught English to middle school students through an self-developed augmented reality (AR) learning module that introduced seasonal characteristics (spring, summer, autumn, winter).

- ▪ Integrated AR-based storytelling and interactive scene exploration to enhance vocabulary learning and student engagement.

Physics Instructor (Online High School Level), Helong High School, Wuhan, Hubei Province, China
Instructor *June* 17, 2020

▪ Taught an online high school physics lesson on friction force (integrated the AR app with slide-based instruction) to a class of 73 students.

▪ Designed complete instructional materials and slides, and developed a self-built augmented reality (AR) application to support physics concept visualization.

PROFESSIONAL WORK EXPERIENCE

University of Florida, USA
Graduate Research Assistant *Aug. 2023– present*

- ▪ Working on the design and development of an educational website called Mflow, using Flow-Based Music Programming to engage children in computer science.
- ▪ Conducting research on how AI assists K–12 computer science education.
- ▪ Leading research sub-projects on AI fairness and LLMs for online learning

Department of Education, Beijing Normal University
Graduate Research Assistant, VR/AR in Education Laboratory Beijing, Sep. 2020 – July 2023

- ▪ Designed and developed AR applications for K–12 education, implementing object recognition and plane detection using C#.
- ▪ Developed and maintained data validation and storage servers using Java.
- ▪ Studied the impact of the AR learning environment on student learning and teacher instruction.

Computer Network Information Center, Chinese Academy of Sciences
Research Assistant, Advanced Interactive Laboratory Beijing, Oct. 2019 – May. 2020

- ▪ Led the visualization research of multi-person and multi-dimensional data interactive sharing based on AR.
- ▪ Developed Augmented Reality software for satellite science using C#, successfully registered it under Software Copyright (Registration No. 2020R11L426768, China).

SERVICE/LEADERSHIP

Reviewer
Academic journals

- ▪ Reviewer of the Computers & Education (C&E) (Since 2024)
- ▪ Reviewer of the Computers and Education Open (Since 2026)
- ▪ Reviewer of the ACM Transactions on Computing Education (TOCE) (Since 2025)
- ▪ Reviewer of the Education and Information Technologies (EIT) (Since 2024)
- ▪ Reviewer of the Frontiers in Education (IEEE) (Since 2025)
- ▪ Reviewer of Scientific Reports (Since 2025)
- ▪ Reviewer of Journal of Computing in Higher Education (Since 2024)
- ▪ Reviewer of Cluster Computing (Since 2025)
- ▪ Humanities and Social Sciences Communications (Since 2025)
- ▪ Sage Open (Since 2025)
- ▪ Reviewer of Discover Education (Since 2025)
- ▪ Reviewer of BMC Psychology (Since 2025)

Conference committee member

- ▪ Educational Advances in Artificial Intelligence (EAAI, 2026)

- The 15th International Learning Analytics & Knowledge Conference (LAK Poster Session) (2025)
- International Society for the Learning Sciences (ISLS) (2025, 2026)
- Special Interest Group for Computer Science Education (SIGCSE TS) (2025, 2026)
- International Conference on Educational Data Mining (EDM) (2024)
- Association for Educational Communications and Technology (AECT) (2024, 2025)
- Association Society for Engineering Education (ASEE) (2024, 2025, 2026)
- IEEE Frontiers in Education (IEEE FIE) (2024, 2025)
- IEEE Global Engineering Education Conference (IEEE EDUCON) (2025)
- American Educational Research Association (AERA) (2024, 2025, 2026)

Student Volunteer

- Special Interest Group for Computer Science Education (SIGCSE TS 2024)
- 27th International Conference on AI in Education (AIED 2026)

MENTORING

Graduate Students *Sep. 2024–Present*

- Mentee: Ms. Yihan Jiang (University of Florida), Ms. Xinyue Jiao (New York University), Ms. Hyunju Oh (University of Florida), Haitao Zhou (Beijing Normal University)
- Resulted in co-authored papers and presentations in renowned journals and conferences.

High School Students *Apr. –Dec. 2025*

- Mentee: Bach Ngo (The Frazer High School).
- Resulted in co-authored papers and presentations in renowned conferences.

LANGUAGES AND SKILLS

- Language: Mandarin (Native) English (Fluent)
- Certifications: Machine Learning, Deep Learning (Coursera Certificate)
- Programming: Python, C#, HTML, Javascript, R, Git, SQL
- Contributed to Software Development:
  - M-Flow (Flow-based programming Platform for Elementary School Computing Education)
  - eSPAC3 (Minecraft Add-ons for Elementary Computing Education)
  - WebART (Web-Based Augmented Reality Learning Resources Authoring Tool)
  - The Wonderful World of Sound 1, 2, 3 (AR application for Elementary School Science)
  - Four Seasons AR Module (AR application for Elementary School English Learning)
  - The National Park (AR application for Elementary School English Learning)
  - Chemical Bonding Visualization (AR application for High School Chemistry)
  - Newton's Three Laws of Motion (AR application for High School Physics)
  - Composition of Forces (AR Application for Middle School Physics)
  - Two-Force Equilibrium (AR Application for Middle School Physics)
  - Why Iron Rusts (AR application for Middle School Science)
  - The Changing Nature of Friction (AR application for High School Physics)

REFERENCES

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